9 - Sub-Orders (Per-Vendor Order Splitting)

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Overview

Today a ReadyCar order is a single indivisible unit: one status, one dispatch gate, one completion event. That does not match reality. A basket bought from three stores becomes three physically separate parcels, collected from three different pickup locations by Bosta, arriving on three different days. Because the system only knows one status, **one slow vendor freezes everybody** — nothing at all can be sent to Bosta until every vendor has reviewed their items, a vendor whose parcel was delivered on Monday is not paid until the last vendor's parcel arrives on Friday, and a customer holding merchandise in their hands sees "Picked up from business" on screen.

This feature splits every order into **sub-orders — one per (vendor, pickup branch)**. The order stays a single financial entity: one price, one payment, one coupon, one loyalty snapshot, one shipping total shown at checkout. But **fulfilment, status, control, money release, and returns all move down to the sub-order**, so each vendor's part of the order lives and finishes on its own schedule.

Goal & Success

- **Goal:** Make the unit of fulfilment control the sub-order, not the order, so no vendor can block another — while the customer still experiences one order, one price, and one payment.
- **Primary user value:**
  - **Customer** — sees exactly what is happening to each part of their order, which store it belongs to, when it arrives, and what each courier will ask them to pay. They can return a delivered part without waiting for the rest.
  - **Vendor** — an unambiguous three-step task list for their own parcel, and money that clears when *their* parcel is delivered.
  - **Admin** — a queue of exactly the sub-orders that need a human, and pickups they can book with photo evidence in front of them.
- **Success looks like:**
  1. A ready vendor's parcel reaches Bosta without waiting for any other vendor.
  2. A vendor's earnings clear from their own delivery date.
  3. A customer can open a return on a delivered sub-order while another is in transit.
  4. The sum of what all couriers collect on a cash order equals the order total exactly — which is **not** true today.
  5. No screen shows a status that contradicts what the customer physically has.

User Types & Permissions

Role	Create	View	Edit	Delete	Notes
Customer	—	Own orders + every sub-order in them, with vendor name	—	Cancel own sub-order **only while no pickup is booked**	Never sees other customers' data, vendor cost, or commission
Vendor	—	Only their own sub-orders and only the items in them	Advance their sub-order through the 3 stages; set item availability + decline reason; upload parcel photos; assign tracking number	Request cancellation (admin approves) **only while no pickup is booked**	Cannot see other vendors on the same order, nor the order total
Admin	Creates a sub-order implicitly by moving an item to another vendor	Everything	Full edit while intervention is open; limited status override after dispatch, with a mandatory reason	Cancel any sub-order; cancel the order (cancels only the not-yet-delivered parts)	The only role that can resolve a declined item

Ownership / role-change rules

- A sub-order belongs to exactly one **(vendor, pickup branch)** pair and never changes owner.
- Moving an item to a different vendor does not transfer the sub-order — it **removes the item from one sub-order and creates or joins another**.
- If a vendor's account is suspended mid-order, their sub-orders freeze into admin intervention; the rest of the order continues.

User Journeys

Entry points

Role	Where they enter
Customer	Profile → My Orders → an order card; also from a partial-delivery notification
Vendor	Vendor dashboard → Orders, opened on the stage tab that needs them
Admin	Admin → Orders (an order), Admin → Pickups (a branch), or the "Needs your attention" queue

Happy path — vendor (the core new journey)

A three-stage progress bar sits at the top of every sub-order the vendor opens.

Stage ① — Confirm products

1. The vendor sees only their own items on this sub-order.

2. Each item has two buttons: **Available** / **Not available**. Pressing either **saves immediately** — there is no separate submit step.
3. **Not available** opens a **mandatory reason** field. The admin needs this text to explain the situation to the customer.
4. A **"All available"** button at the top confirms every item at once.
5. When no item is left unreviewed, the **"Confirm product availability"** button unlocks.
6. Pressing it — **with every item marked available** — creates the Bosta delivery automatically and advances to stage ②.
7. Pressing it with **any item declined** does not advance. The sub-order freezes and moves to the admin's intervention queue (see *Alternate paths*).

Stage ② — Assign tracking number

The screen shows one of two options, decided by whether Smart Label is switched on for the site:

- **Smart Label on** — "Scan the label" (or type the number).
- **Smart Label off** — "Download the shipment PDF and stick it on the parcel." The PDF is available because the delivery was created at the end of stage ①.

Completing either advances to stage ③.

Stage ③ — Ready for pickup

1. The vendor uploads **1 to 5 photos** of the parcel. At least one is required.
2. Pressing **"Confirm ready for pickup"** marks the sub-order ready.
3. It now appears in the admin's Pickups module for its branch, with its photos and tracking number attached.

From here Bosta drives the status: pickup scheduled → picked up → in transit → out for delivery → delivered.

Happy path — customer

1. Opens an order and sees one header (reference, total, and a progress counter) with a card per sub-order beneath it.
2. Each card carries the **vendor name and branch**, the items in it, its own status, its own tracking, and — on cash orders — **the exact amount that courier will ask for**.
3. A push/WhatsApp notification arrives per partial delivery, naming the vendor and how many parcels remain.
4. When a card reaches *Delivered*, a **Request return** button appears on that card only, with its own return window counted from that card's delivery date.

Happy path — admin

1. Opens an order and sees a card per sub-order, each showing its stage, tracking, parcel photos, commission, and action buttons.

2. Opens Pickups and sees, per branch, the list of sub-orders ready today with their photos — reviews them, then books the courier for the parcels they accept.

Alternate paths

Situation	What happens
Any item declined (partial or all)	Sub-order freezes; enters the admin queue with the vendor's reason. The vendor cannot proceed.
Admin resolves — replace from the same vendor	The item returns to that sub-order at stage ①; the vendor must confirm it again.
Admin resolves — replace from a different vendor	The item leaves this sub-order and **joins or creates the other vendor's sub-order**, starting at stage ① there. Shipping is re-quoted.
Admin resolves — remove the item	The item's value is refunded to the customer's wallet (or deducted from the COD if not yet dispatched).
Admin resolves — cancel the sub-order	Its share is refunded to the wallet; its loyalty points are returned; the rest of the order continues.
Vendor misses the deadline	The sub-order is flagged *overdue* in the admin list; the vendor receives a reminder before the deadline expires. It is **never auto-cancelled**.
Sub-order goes to exception	Only that card turns red. The order keeps progressing on the other sub-orders.
Sub-order returned to the vendor	Only that sub-order closes as returned; the rest is unaffected.

Exit / confirmation

- **Vendor** — the progress bar shows all three stages complete and the sub-order moves to the "On the way" tab.
- **Customer** — every card reads *Delivered*, the header reads *Completed*, and the return window on each card counts down from its own delivery date.
- **Admin** — the sub-order leaves the intervention and pickup queues.

UX & Layout

Customer — order detail

One order header, then **a card per sub-order**.



- **Primary actions:** *Request return* and *Cancel* live **inside the card they act on** — never in the header — so it is impossible to cancel the wrong parcel.
- **Cancel** is only rendered while no pickup is booked for that sub-order.
- **Mobile:** cards stack full-width. Tabs were rejected — the customer must be able to see the whole picture in one scroll.

Vendor — sub-order detail

A three-stage progress bar pinned at the top, the active stage expanded and the others collapsed. The single primary action for the current stage is a full-width button at the bottom of the stage, disabled until its requirements are met, with the reason it is disabled written next to it ("Upload at least one photo").

Vendor — orders list: tabs by stage with counts — *Needs review (3) · Needs tracking (1) · Needs photos (2) · On the way · Delivered.* The vendor lands on the leftmost non-empty tab, so the next thing to do is always the first thing they see.

Admin — order detail

The existing vendor groups become **sub-order cards**, each carrying: the stage indicator, status, item count and totals, commission, parcel photo thumbnails, the AWB link, and its own action buttons. An orange banner at the top of the page counts the sub-orders waiting for intervention and links to the first one.

Admin — pickups

Per branch, above the *Book courier* button, the list of sub-orders ready today — each with its reference, photo thumbnails, and tracking number — plus a muted line for those not yet ready. The button states how many parcels it is booking.

Navigation placement

No new top-level navigation. Sub-orders are always reached through the order they belong to, from all three sides.

Cognitive-load notes

- The vendor is never shown a form with more than one decision at a time.
- Buttons save on press; there is no draft state to lose and no submit step to forget.
- The customer sees a counter ("1 of 3 delivered") rather than a single status word, because one word cannot describe three parcels honestly.

States

- **Empty** — a vendor with no sub-orders in the active tab sees the tab's own message ("Nothing waiting for tracking numbers") rather than a generic empty state.
- **Loading** — cards render as skeletons in place; the order header and totals appear first, since they are already known.
- **Success** — advancing a stage animates the progress bar forward and confirms inline. No modal, no page reload.
- **Error**
  - *Bosta refused to create the shipment* — the sub-order stays at stage ① with the provider's reason shown, and the vendor can retry. Nothing else in the order is affected.
  - *Photo upload failed* — the stage stays open and already-uploaded photos survive.
  - *Sub-order in exception* — the customer's card turns red, states the reason in plain language, and offers **Contact support**, which opens a ticket already linked to that sub-order.
- **Partial / missing data**
  - A sub-order with no tracking number yet shows "Preparing" instead of a blank field.
  - A legacy order migrated without parcel photos shows "No photos (order predates this feature)" rather than an empty gallery.
  - A vendor whose branch has no Bosta location shows an admin-only warning that this pickup must be arranged by hand.

Business Rules

Status transitions

Sub-order

Status	Meaning	Entered by
`pending_review`	Vendor has unreviewed items	Order placement
`awaiting_tracking`	All items available; Bosta delivery created	Vendor presses *Confirm product availability*

Status	Meaning	Entered by
`awaiting_handover`	Tracking number assigned; photos required	Vendor scans label / downloads AWB
`ready_for_pickup`	Photos uploaded; visible in Pickups	Vendor presses *Confirm ready for pickup*
`pickup_scheduled` → `picked_up` → `in_transit` → `out_for_delivery` → `delivered`	—	**Bosta only**
`needs_admin`	Any item declined, or vendor suspended	Automatic
`cancelled`	—	Vendor request + admin approval, admin, or customer
`returned`	Parcel came back to the vendor	Bosta
`exception`	Bosta problem state	Bosta

Order (derived, never set directly)

`pending` → `processing` → `partially_shipped` → `partially_delivered` → `completed`, plus `cancelled` and `exception`. Two new values — `partially_shipped` and `partially_delivered` — exist so the admin can filter for them. Every screen renders them alongside a counter: "*1 of 3 delivered.*"

- `completed` **only** when every sub-order is `delivered`.
- A sub-order in `exception`, `returned`, or `cancelled` is **excluded** from the rollup rather than poisoning it — the order keeps progressing on the rest.
- The order is `cancelled` only when **every** sub-order is cancelled.

Validations

1. **Stage ① cannot complete with a declined item.** Any decline routes to `needs_admin`.
2. **A decline requires a written reason.** The admin uses that text with the customer.
3. **Stage ③ requires at least one photo, maximum five.**
4. **Cancellation requires that no pickup is booked for that sub-order.** Once booked, the cancel action is hidden from both the vendor and the customer.
5. **Financial allocation freezes at first dispatch.** The moment any sub-order in the order reaches `awaiting_tracking`, the split of shipping, coupon, and loyalty discount across sub-orders is frozen. Later cancellations refund their own frozen share and never re-price the surviving sub-orders — because Bosta offers no way to change a COD amount after a delivery is created.
6. **The admin cannot edit an order after the allocation freezes**, except to cancel a sub-order. Intervention must be resolved before the first vendor dispatches.
7. **A coupon is re-validated only on a customer-initiated cancellation.** If the remaining merchandise falls below the coupon's minimum, the coupon is dropped and the refund is computed without it. A cancellation caused by a vendor or the admin never costs the customer their discount.

8. **COD per sub-order** = its merchandise + its shipping share − its coupon share − its loyalty share. The sum across all sub-orders equals `total_price` exactly.
9. **Shipping is quoted per sub-order** (split legs) and presented to the customer as a single total at checkout.
10. **Admin status override after dispatch** is limited to closing a sub-order (delivered / returned / cancelled) and always requires a written reason recorded in the history.
11. **An order-total change on a prepaid order:** a decrease is refunded to the wallet immediately; an increase is added to that sub-order's COD and collected on delivery.

Notifications

Recipient	Trigger	Content
Customer	A sub-order is delivered	Names the vendor and how many parcels remain
Customer	A sub-order is out for delivery (cash orders)	The exact amount that courier will collect
Customer	A sub-order enters exception	Plain-language reason plus a support link
Vendor	Deadline approaching	Which stage they are stuck on and how long is left
Vendor	Admin resolves an intervention on their sub-order	What changed and that action is needed again
Admin	Any decline, overdue sub-order, or exception	Enters the single "Needs your attention" queue with its reason

The existing order-confirmation message at placement is unchanged — one message, one order.

On cancel / delete / restore

- **Vendor-requested cancellation** — an admin approval step. On approval the system terminates the Bosta delivery, releases the smart label, refunds that sub-order's frozen share to the wallet, returns its loyalty points, and marks it `cancelled`.
- **Customer cancellation** — same effects, plus coupon re-validation (rule 7).
- **Cancelling the whole order** — cancels only the sub-orders that are not yet delivered. Delivered ones stay; the customer uses the normal return flow for those. The order is then shown as partially cancelled.
- **Deleting an order** — unchanged, admin only, and only before dispatch.

Concurrent / conflicting actions

- **A Bosta webhook and an admin override race.** Bosta wins for forward progress; an admin's manual close wins only if it is more advanced than what Bosta reports, and is recorded with its reason either way.
- **Two admins resolving the same intervention.** The first resolution wins; the second is refused with the sub-order's current state shown.
- **A vendor advancing a stage while an admin is editing.** The freeze rule settles it: once any sub-order dispatches, editing is closed, so the two cannot overlap on money.

- **A pickup being booked while a customer presses cancel.** The booking wins; the cancel is refused and the customer is told the parcel is already scheduled.

Edge Cases

Rare or invalid operations

- One vendor shipping from two branches → **two independent sub-orders**, each with its own stepper, tracking number, photos, earnings row, and return window. The customer sees the vendor's name twice with different branches.
- A vendor declines every item → same path as a partial decline: freeze and hand to the admin, who may source a replacement from another vendor.
- An admin replaces an item with one from a vendor whose sub-order has already been dispatched → a **new** sub-order is created for that vendor rather than reopening the dispatched one.
- Bosta rejects the delivery at stage ① → the vendor stays at stage ① with the reason and can retry; no other sub-order is touched.
- A vendor stalls for weeks at stage ② → flagged overdue, never auto-cancelled; a live Bosta delivery exists, so only an admin cancellation may terminate it.

Abuse scenarios

- *Coupon erosion.* A customer cancels sub-orders until the remainder falls below the coupon minimum → re-validation on customer-initiated cancellation drops the coupon.
- *Free-shipping erosion.* Same mechanism, same guard.
- *Vendor gaming photos.* Photos are timestamped and attributed; the admin can refuse a parcel at the pickup stage.
- *Vendor cancelling to dodge a bad-margin order.* Cancellation is a request, not an action, and every request is recorded against the vendor.

Data-consistency risks

- *COD drift.* Bosta exposes no way to update a delivery's COD after creation (`bostaService` offers only `createDelivery` and `terminateDelivery`). The freeze rule is the mitigation — see the Decision Log.
- *Split rounding.* Shares are allocated at the smallest currency unit using the existing largest-remainder method, so the parts sum to the total with no drift.
- *Migration ambiguity.* A legacy order whose shipments do not record item ids cannot be split with confidence and is migrated as a single sub-order containing everything.
- *Duplicate earnings.* An earnings row is keyed per sub-order and created once, on that sub-order's own delivery.

Permission conflicts

- A vendor must never see another vendor's items, totals, or the order total. Sub-orders make this structural rather than a filter applied at read time.
- A customer must not see commission or vendor cost anywhere on a sub-order card.

- A vendor must not see the customer's full address before their sub-order reaches stage ② — the current rule is unchanged.

Decision Log

Sub-orders are embedded in the order, not separate orders

- **Decision:** One order document keeps one price, one payment, one coupon, and one loyalty snapshot. Sub-orders are a structure inside it.
- **Reason:** Everything financial is already order-level and singular. Splitting into separate order documents would force one payment, one coupon, and one loyalty spend to be distributed across N documents — a far harder problem than the one being solved, and one that risks real money.
- **Alternatives considered:** True separate orders per vendor (rejected: distributed money handling); display-only grouping (rejected: does not remove the dispatch gate).
- **Impact:** The customer's single-order experience is preserved for free. Reporting and refunds keep working against one document.

The sub-order is (vendor + branch), not vendor

- **Decision:** A vendor shipping from two branches gets two sub-orders.
- **Reason:** Each branch is a physically separate parcel with its own pickup, tracking number, photo, and delivery date. One stepper cannot honestly represent two parcels.
- **Alternatives considered:** One sub-order per vendor holding several shipments (rejected: the stepper becomes ambiguous and the return window undefined); forcing the vendor to consolidate into one branch (rejected: blocks vendors with distributed stock).
- **Impact:** Earnings and returns must also become per sub-order — accepted deliberately (next entry).

Earnings and returns move to the sub-order

- **Decision:** Both the earnings ledger and return requests are keyed per sub-order rather than per (order, vendor).
- **Reason:** With per-branch sub-orders, a single per-vendor row cannot answer "which delivery date?" when two branches arrive four days apart. Keeping them per vendor would reintroduce exactly the blocking this feature exists to remove — a vendor's first branch waiting on their second.
- **Alternatives considered:** Keeping the money per vendor and clearing on the last delivery (rejected: the same blocking, one level down).
- **Impact:** The largest data-migration cost in this feature, and the reason existing orders must be converted rather than left on the old model.

The financial allocation freezes at first dispatch

- **Decision:** When the first sub-order in an order reaches stage ②, the split of shipping, coupon, and loyalty across sub-orders is frozen. Later cancellations refund their own frozen share and never re-price the survivors. The admin cannot edit the order after that point, only cancel.

- **Reason:** Bosta has no API to change a COD amount once a delivery exists. Without the freeze, cancelling one sub-order would re-allocate shipping and coupon shares onto a sibling whose COD is already locked at Bosta, and the courier would collect the wrong amount — hundreds of pounds on a typical basket.
- **Alternatives considered:** Creating the delivery only at stage ③ (rejected: the vendor would have to stick the label on after photographing the parcel); terminating and recreating the delivery on every change (rejected: a new tracking number each time, forcing the vendor to re-print or re-scan).
- **Impact:** Interventions must be resolved **before** the first vendor dispatches. In the rare case where a later cancellation shifts the true shipping burden, the store absorbs the difference rather than surprising the customer at the door.

Coupons are re-validated only when the customer cancels

- **Decision:** A customer-initiated cancellation that drops the order below a coupon's minimum drops the coupon. A vendor- or admin-initiated cancellation never does.
- **Reason:** A customer must not lose a discount because a vendor ran out of stock, and the store must not fund an unearned discount when the customer chose to shrink the order.
- **Alternatives considered:** Always re-validate (rejected: punishes customers for vendor failures — a guaranteed complaint); never re-validate (rejected: an exploitable gap once customers can cancel parts).
- **Impact:** The refund calculation must know **who** initiated the cancellation.

COD carries its share of shipping and discounts

- **Decision:** Each sub-order's cash-on-delivery amount is its merchandise plus its shipping share, minus its coupon and loyalty shares. The sum equals the order total.
- **Reason:** Today the couriers collect merchandise only — so the store never collects shipping on cash orders, and the customer is charged the value of a coupon they were granted. It is invisible now because there is one blended number; per-sub-order cards make it visible on every order.
- **Alternatives considered:** Putting all shipping on the first parcel (rejected: breaks if that parcel is returned); leaving the calculation as-is (rejected: a known revenue leak that sub-orders would put in front of every customer).
- **Impact:** Fixes an existing defect that is independent of this feature. The allocation reuses the share arithmetic already written for refunds.

The vendor works a three-stage wizard, not a status dropdown

- **Decision:** Confirm products → assign tracking → confirm ready, with the availability and photo requirements as gates inside the stages rather than statuses of their own.
- **Reason:** The admin only needs one answer from the vendor — "can I put this on today's pickup?" — and all three stages serve that one question. A longer status list would clutter every dashboard without answering anything more.
- **Alternatives considered:** Seven separate statuses (rejected: dashboard noise); reusing the existing order statuses with checkmarks (rejected: gives the vendor no sense of a path to follow).
- **Impact:** Stage completion is the natural place for the deadline to be measured.

Item review saves on press

- **Decision:** Availability buttons write immediately; there is no submit step. A decline opens a mandatory reason. An "All available" button confirms everything at once.
- **Reason:** The current toggle-then-submit pattern loses work and leaves the admin guessing why an item was declined.
- **Alternatives considered:** Keeping the submit button (rejected: the source of the complaint); adding a "partial quantity" option (rejected for now — it opens a whole price-difference workflow; recorded as an open question).
- **Impact:** The admin's "vendor finished reviewing" signal becomes simply "no item is left unreviewed."

Existing orders are all migrated

- **Decision:** Every live order converts to sub-orders at release, with each sub-order's status derived from its actual shipment state.
- **Reason:** Two order shapes in one admin list is a support burden that outlasts the release and makes every bug report ambiguous.
- **Alternatives considered:** New orders only (rejected: two UIs indefinitely); migrating only undispatched orders (rejected: the same split-brain, smaller).
- **Impact:** Migration must be reversible and must be rehearsed against a copy of live data before release.

Open Questions

□ What is the default deadline value, and is it measured from order placement or from when the sub-order first becomes actionable to the vendor?

□ Should the deadline be overridable per vendor (a vendor whose goods are made to order needs longer), or is one global number enough for now?

□ "Partial quantity available" (vendor has 1 of 2) was deferred. Is it needed in v1, or does the admin-intervention path cover it well enough?

□ Exactly which Bosta exception reasons are shown to the customer verbatim, and which are replaced with a neutral message?

□ Do parcel photos need a retention limit, and who may delete them after delivery?

□ Should the admin be able to reject parcel photos and push a sub-order back to stage ③, or only exclude it from today's pickup?

□ Does the customer's Cancel button need a reason field, and should repeated cancellations be rate-limited?

□ Should the vendor see the customer's phone number at stage ②, or stay on the current rule?

□ How are family and friend orders presented — does the recipient see per-sub-order cards, or only the payer?

□ What happens to a sub-order whose vendor is hard-deleted (feature 1) while the parcel is in transit?